RevPlay: Full-Stack Music Streaming Platform

A comprehensive monolithic web application built with Spring Boot and Angular for seamless music discovery, playback, and artist engagement

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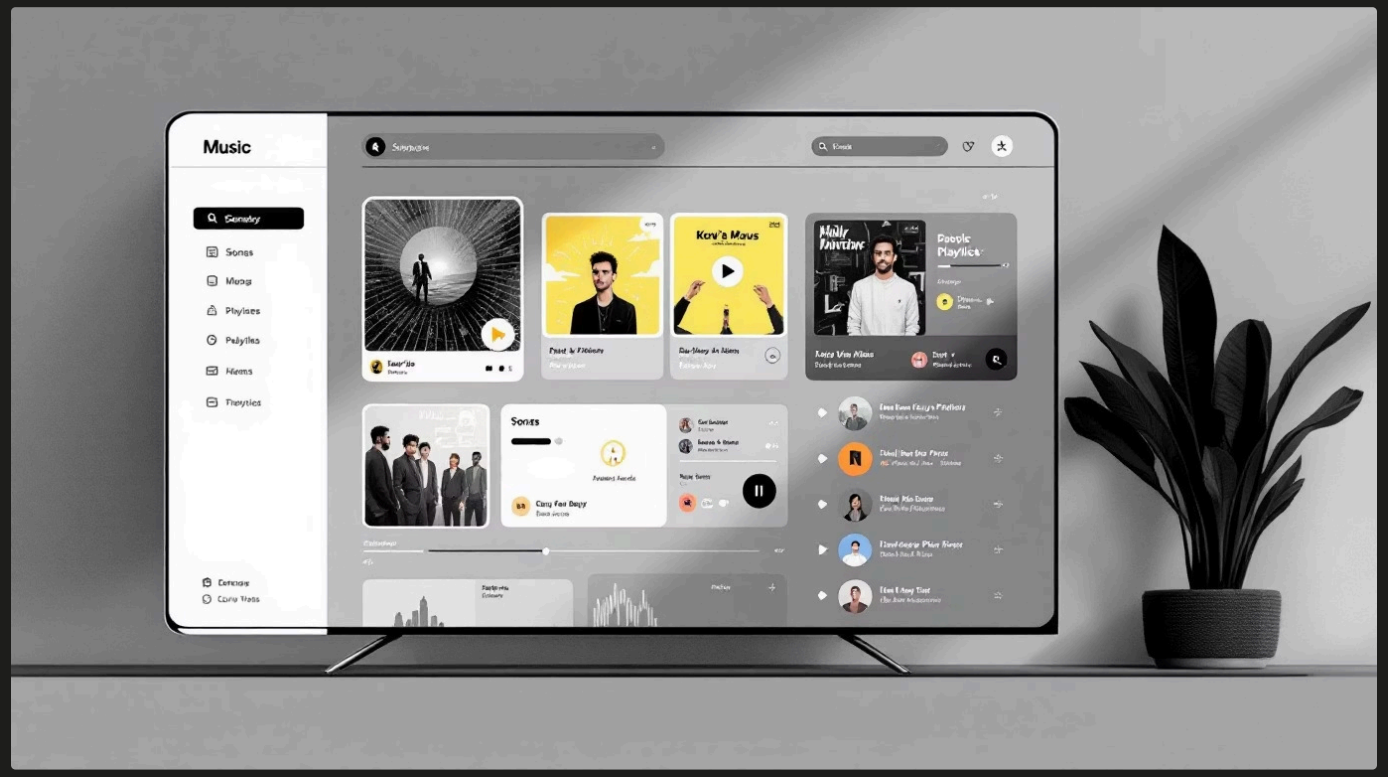
Application Overview



What is RevPlay?

RevPlay is a full-stack monolithic music streaming web application that provides users with access to an extensive library of songs, artists, playlists, albums, and podcasts. The platform features responsive design, role-based access control, integrated music player, and real-time analytics.

Both listeners and artists can create profiles, with musicians gaining additional tools to showcase their work and track engagement through comprehensive statistics.

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Technology Stack

Backend Framework

Spring Boot

RESTful APIs and service layer implementation

Frontend Framework

Angular

Responsive UI components and routing

Database

SQL

Entity relationship management

Security

Log4J

Authentication and logging

Version Control

Git

Code repository management

Build Tool

Maven , Java

Dependency management

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Core User Features

01

Authentication & Profile Management

Secure registration and login with email/username credentials. Users can view and edit profile details including display name, bio, and profile picture.

02

Music Discovery & Search

Browse extensive music library by categories, songs, artists, and albums. Powerful search functionality for finding songs, playlists, and artists instantly.

03

Integrated Music Player

Play songs with full controls: play, pause, volume adjustment modes for personalised listening experiences.

04

Favorites & Collections

Mark songs as favorites for quick access. Create and manage playlists with custom names, descriptions, and privacy settings (public or private).

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Music Player Capabilities

Essential Playback Features

- Play, pause controls for seamless navigation
- Progress bar with seek functionality for precise positioning
- Volume control with mute toggle
- Repeat mode options: off, repeat one song, or repeat all playlist
- Currently playing song display with album art
- Playback queue management for upcoming tracks

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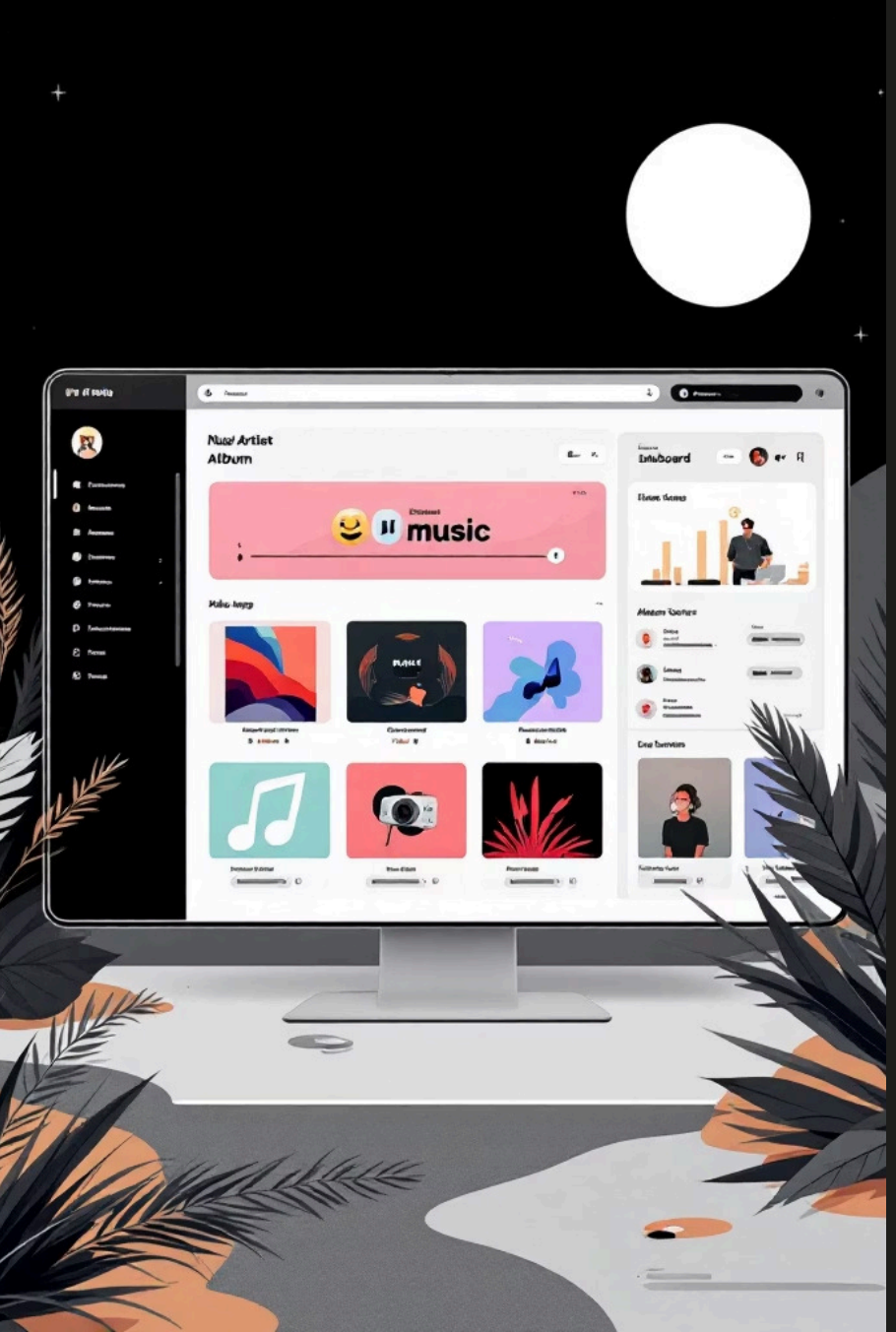

Artist Portal Features

Artist Profile Management

Create comprehensive artist profiles with name, gender, role.

Music Upload & Management

Upload songs with complete details: title, genre, audio file, and album artwork. Create albums with descriptions, and cover art. Organise songs into albums and update metadata.

Content Visibility Control

Set song visibility to public or unlisted. Delete songs and albums (when empty). Manage which content appears in the main library versus restricted access.

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Analytics & Insights

Real-Time Statistics Dashboard

Artists gain access to comprehensive analytics showing their music's performance and audience engagement.

- Total songs uploaded and total plays accumulated
- Individual song play counts and popularity rankings
- Users who favorited specific songs
- Daily, weekly, and monthly listening trends
- Top listeners and engagement metrics

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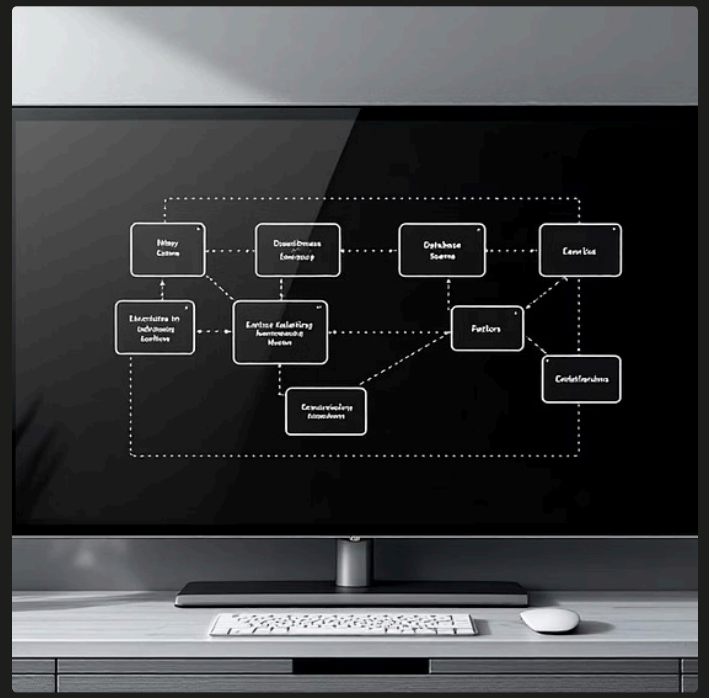
Key Technical Specifications

Responsive Design

Adaptive interface that works seamlessly across desktop, tablet, and mobile devices with consistent user experience

Role-Based Access

Secure authentication system with distinct permissions for regular users and verified artists with extended capabilities



Database Architecture

Well-designed ERD with relationships between users, songs, artists, albums, playlists, and analytics data

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Implementation Highlights

Spring Boot Backend

RESTful API endpoints, service layer, repository pattern, and dependency injection for robust backend operations

Angular Frontend

Component-based architecture, reactive forms, HTTP client, and routing for dynamic user interface

SQL Database

Normalized schema with efficient queries, indexing, and relationships between all entities and data structures

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Project Completion & Deliverables

Definition of Done

Working Web Application

Complete, functional music streaming platform with all core features implemented and tested

Code Repository

Organised source code with Entity Relationship Diagram (ERD) and Application Architecture documentation

README Documentation

Comprehensive README.md file explaining installation, configuration, and usage instructions

Testing Artifacts

JUnit4 tests (v6) verifying functionality with test coverage reports and results

Ready for Production

RevPlay delivers a complete full-stack solution with robust authentication, music streaming capabilities, artist tools, and analytics – everything needed for a professional music platform.

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conclusion

RevPlay stands as a comprehensive, full-stack music streaming platform, meticulously designed with robust architecture to deliver seamless performance and essential functionalities. It encompasses everything from secure authentication to a fully operational music streaming experience, complete with artist tools and comprehensive documentation.

Beyond its technical foundation, RevPlay empowers artists with intuitive tools for creation, distribution, and audience engagement, while offering real-time analytics for strategic decision-making. This production-ready solution is engineered for immediate deployment and poised to make a significant impact in the digital music landscape.

RevPlay is not just a platform; it's a meticulously engineered solution poised to orchestrate the future of music streaming.

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